UTKARSH SHARMA

Software Systems & Data Science · Undergraduate Builder

GitHub: github.com/shutkarshX
LinkedIn: in/utkarsh-sharma
Production: tein.onrender.com · Greater Noida, India

EDUCATION

Academic Foundation

Noida Institute of Engineering & Technology (NIET)

Aug 2025 — Apr 2029

Bachelor of Technology (B.Tech) in Computer Science & Engineering — Greater Noida, India

- **Specialization:** Data Science & Engineering Systems.
- **Relevant Coursework:** Data Structures & Algorithms, Systems Programming, Database Management Systems, Object-Oriented Software Design (Java), Computer Networks, Statistical Data Analysis.

EXPERIENCE

Industry Internship

Prodigy InfoTech

Sep 2026 — Oct 2026

Data Science Intern · Remote — Python · Pandas · Scikit-Learn · Seaborn

- Completed an intensive 5-milestone curriculum delivering applied machine learning models, exploratory data analysis, and NLP classification pipelines.
- Built and tuned **Decision Tree classifiers** on customer demographic datasets to model subscription conversions, optimizing precision, recall, and decision split thresholds.
- Conducted in-depth **Exploratory Data Analysis (EDA)** on demographic and Titanic survival datasets using Pandas and Seaborn to identify correlations and feature significance.
- Engineered text preprocessing, tokenization, and sentiment analysis pipelines in Python to categorize social media public sentiment patterns.

KEY ENGINEERING PROJECTS

Production Platforms, AI & Native Systems

TEIN — Attendance Intelligence & Campus Hub Platform [tein.onrender.com ↗] [repo ↗]

Python · Flask · Docker · PWA · Render

- Engineered and deployed a mobile-first university platform live on Render solving attendance compliance and student society discoverability across semester cycles.
- Formulated **Bunk & Recovery algorithms** calculating exact safe class skip margins and automated recovery schedules to preserve institutional 75% attendance policy.
- Built a forward-looking **What-If Simulator** enabling students to model hypothetical absence impacts across upcoming calendar dates prior to taking leave.
- Implemented offline-first **PWA caching architecture** (`/sw.js`) and an integrated directory indexing 20+ active campus clubs and technical societies.

RESQ-AI — Intelligent Disaster Response Command Center [repo ↗]

FastAPI · React · Gemini AI · Leaflet GIS

- Architected an emergency coordination dashboard integrating **Google Gemini AI** to triage citizen distress calls and extract geospatial coordinates into prioritized dispatch tickets.
- Rendered real-time interactive command maps with **Leaflet GIS** plotting active incident pins, rescue convoy routes, and relief shelter capacity.
- Engineered asynchronous FastAPI backend with strict Pydantic validation schemas, non-blocking I/O event loops, and auto-generated Swagger documentation.

Layer-Flow — Transparent Screen Annotation & Desktop Canvas [repo ↗]

Rust · Tauri 2 · React · Win32 API

- Developed a native Windows transparent desktop overlay in Rust and Tauri 2 with sub-50MB memory footprint and zero UI latency, avoiding Chromium runtime overhead.
- Integrated low-level **Win32 API hooks** for global system hotkeys, click-through mode toggling (`WS_EX_TRANSPARENT`), and multi-monitor pen annotation.

Speedo — Coursework Progression & Adaptive Video Controller [repo ↗]

JavaScript · Chrome MV3 · DOM Automation

- Authored a Manifest V3 Chrome extension to automate academic video progression, bypass playback speed restrictions, and streamline digital coursework.
- Implemented a **CSP-safe Main World injection bridge** adhering to strict Manifest V3 Content Security Policies without security sandbox violations.
- Constructed an automated observer dismissing inactivity modals to maintain uninterrupted coursework progression.

TECHNICAL CAPABILITIES

Engineering Stack

Languages:	Python, Rust, JavaScript, TypeScript, C (C99), Java, SQL, R, HTML/CSS
Frameworks & Libs:	FastAPI, Flask, React, Vite, Tauri 2, Pandas, Scikit-Learn, Seaborn, NumPy, Leaflet
Systems & Cloud:	Docker, Linux / Bash, Progressive Web Apps (PWA), Win32 API, Render Cloud, Git / GitHub Actions
Testing & Tools:	Pytest (190+ automated tests across repos), REST APIs, Asyncio, Chrome MV3, VS Code